Cardiovascular System (CARD) · NPLEX-P1-CARD-10

CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs

Work each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Spread the two brain dumps for a competency across different days rather than doing them back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-10-01-C01				
NPLEX-P1-CARD-10-01-C02				
NPLEX-P1-CARD-10-01-C03				
NPLEX-P1-CARD-10-01-C04				
NPLEX-P1-CARD-10-01-C05				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C01-BD1

Draw the lymphatic capillary and how lymph forms

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-BD1 Date: _____

Without looking at your notes, draw a lymphatic capillary sitting in the interstitium beside a blood capillary. Label the blind end, the overlapping endothelial cells, the flap that acts as a valve, the basal lamina, and the anchoring filaments, and write beside each label what it does. Then write the four Starling forces, mark which two push fluid out and which two hold it in, and write one sentence on why a small net filtration leaves the interstitium with both fluid and protein in it. Finish by writing in order the five steps that take fluid from net capillary filtration to lymph moving on into a valved collecting vessel, and write beside the first and last steps what opens the vessel and what closes it.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C01-BD2

What stops working when protein cannot get back

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-BD2 Date: _____

A patient's lymphatic return from one leg is blocked while capillary pressure, plasma albumin, and capillary permeability all stay normal. Write what happens to the protein concentration of the interstitial fluid in that leg and why no other route can clear it, then write what that does to interstitial oncotic pressure, to the gradient that normally holds fluid inside vessels, and to net filtration. Next write the protein concentration of lymph from most tissues, from the liver, and from the intestine, and explain from the structure of hepatic sinusoids why the liver figure is the highest. Then a second patient has a thoracic duct that has been draining externally for ten days. Write which fat-soluble components, which plasma component, and which cells are being lost, and explain from the size of chylomicrons why dietary long-chain fat leaves by this route while medium-chain fatty acids do not. Finish by naming the tissues that have no lymphatic capillaries at all.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-10-01-C01-RT01

Rebuild the steps that form lymph

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C01 Activity: NPLEX-P1-CARD-10-01-C01-RT01 Date: _____

Rebuild from memory the five-step sequence that runs from net capillary filtration to lymph entering a valved collecting lymphatic, one step per line and in order. Beside the step where the vessel opens, draw the anchoring filaments and show with arrows which way the tissue pulls. Beside the step where the vessel closes, draw the endothelial flaps and show which side the pressure comes from. Underneath, write the resting lymph flow per hour, the share of it carried by the thoracic duct, and the daily volume, and write one sentence saying why that daily volume matches net filtration.

Rebuild from memory: Formation of lymph

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C02-BD1

Draw the thoracic duct from end to end

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C02 Activity: NPLEX-P1-CARD-10-01-C02-BD1 Date: _____

Draw the vertebral column in outline from L2 to C7 and lay the thoracic duct on it from memory. Mark and label the cisterna chyli with its vertebral level, the two lumbar trunks and the intestinal trunk that feed it, the aortic hiatus with its level, the level at which the duct crosses from right to left, and the vein junction where it ends. Write beside the posterior mediastinal part which structure lies in front of the duct, which vessel lies on each side of it, and what the terminal valve does. Then draw the right lymphatic duct, name the three trunks that form it, and shade the four body regions it drains. Finish by writing which lung lobe drains against the pattern you have just shaded and where its lymph goes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C02-BD2

Work backward from the node to the primary site

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C02 Activity: NPLEX-P1-CARD-10-01-C02-BD2 Date: _____

Five patients each present with one enlarged node group: left supraclavicular, para-aortic near the renal vessels, superficial inguinal, axillary, and popliteal. For each one write which regions drain there and name one primary site that would explain the finding. Then explain three of those patterns from development rather than from memory: why a testicular tumour reaches para-aortic nodes while a scrotal skin lesion reaches superficial inguinal nodes, why the anal canal drains to two different node groups depending on which side of the pectinate line a lesion sits, and why an abdominal carcinoma can surface first in a node at the root of the left side of the neck. Finish by writing which single node group drains most of the breast, which group drains its medial part, and why a lesion of the medial hand can enlarge a node at the elbow before any node in the axilla.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-10-01-C02-PR01

Predict the effect of a block low in the thoracic duct

A retroperitoneal lymphoma compresses the thoracic duct just above the cisterna chyli. Predict each item below, state the direction of the change or name the structures involved, and give the anatomical reason in one line.

What changes	Up, down, or no change	Why, in one or two lines
What happens to lymph outflow from both lower limbs and the pelvis		
What happens to lymph outflow from the abdominal viscera, including intestinal chyle		
Which regions of the left side of the body still drain normally, and through which vessels they reach the veins		
Which regions of the right side of the body still drain normally, and through which vessel they reach the veins		
What happens to the pressure inside the duct below the obstruction		
What happens to the valve leaflets of the dilated collecting vessels below the obstruction, and what that does to forward flow		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C03-BD1

Lay out the whole lymphatic pump

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-BD1 Date: _____

Lymph starts in an interstitium below atmospheric pressure and ends in veins above it, in the legs against gravity. Write the pressure at each end, then write why valves are what make any compression useful. Draw two lymphangions in series with their valves labelled, and write out the six steps of one pump cycle from filling to refilling. Beside the cycle write what stretch from filling does to the rate and force of contraction, and name the cardiac property this resembles. Then list the three influences on that smooth muscle from the notes and the direction each one pushes its activity. Finish by listing the extrinsic pumps, writing the fold rise in flow with exercise, and explaining step by step how one inspiration moves lymph up the duct.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C03-BD2

Why the pump runs out of room

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-BD2 Date: _____

Draw interstitial fluid pressure on the horizontal axis from -6 mmHg to +4 mmHg and lymph flow on the vertical axis, and sketch the relationship. Mark the normal subcutaneous value, write why it is negative, mark where flow has risen about twentyfold, and mark where the curve flattens. Write two sentences on why flow stops rising past that point even though the tissue keeps swelling, and write what accumulates once filtration exceeds that ceiling. Next write the three safety factors that oppose edema with the pressure each is worth and the total, and write what each one depends on. Then remove the regional nodes and vessels from a limb: write which safety factors are lost immediately, write what happens to interstitial protein and therefore to interstitial oncotic pressure, and explain why that limb swells even with a normal capillary pressure. Finish with two patients whose lymph outflow is slowed at the far end rather than the near end, one with right heart failure and one with superior vena cava obstruction, and write for each why filtration rises at the same time as outflow falls.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-10-01-C03-RT01

Rebuild the lymphangion pump cycle

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C03 Activity: NPLEX-P1-CARD-10-01-C03-RT01 Date: _____

Rebuild from memory the six-step cycle of one lymphangion, one step per line and in order, starting with lymph entering through the open upstream valve. Draw the segment at each of the six steps as a small sketch and shade which valve is open and which is closed. Beside the contraction step write what triggered it and what the trigger was before that. Beside the ejection step write where the ejected lymph goes and what it does to the next segment. Underneath, write the pressure an obstructed thoracic duct can generate, and write what happens to the leaflets and to forward flow if a chronically obstructed vessel dilates.

Rebuild from memory: The lymphangion pump cycle

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C04-BD1

Build the lymphoid organs from their compartments

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C04 Activity: NPLEX-P1-CARD-10-01-C04-BD1 Date: _____

Separate the primary from the secondary lymphoid organs, write which lymphocyte matures in each primary organ, and write what each secondary organ samples. Then draw four diagrams from memory and label every compartment named below. For a lymph node: capsule, subcapsular sinus, cortical follicle with germinal center, paracortex, medullary cords, hilum, and the afferent and efferent vessels, with arrows for the direction lymph travels. For the spleen: central arteriole, periarteriolar lymphoid sheath, follicle, marginal zone, and red pulp with its cords and sinusoidal slits. For a thymic lobule: cortex, medulla, Hassall corpuscle, and the corticomedullary junction, and write the two things the thymus does not have. For a Peyer patch: epithelium with its M cell, follicle, and the direction antigen travels. Beside each compartment write the main cell type it holds and the one job it does. Finish by writing the eight-step sequence of T-cell maturation in the thymus from marrow precursor to a naive T cell leaving, and mark which steps happen in the cortex and which in the medulla.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C04-BD2

Read a biopsy and a blood film back to a compartment

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C04 Activity: NPLEX-P1-CARD-10-01-C04-BD2 Date: _____

Three children have node biopsies. The first has follicles and germinal centers present but an empty paracortex. The second has a normal paracortex but no follicles, no germinal centers, and no plasma cells in the medullary cords. The third has a greatly enlarged paracortex with normal follicles. For each child write which cell line is missing or responding, write what the same defect would do to the spleen's periarteriolar sheath, marginal zone, and follicles, and name the condition or the kind of infection that fits. Explain for the second child why serum antibody of every class is low. Next write what a T-cell receptor must do in the thymic cortex to survive and what it must avoid doing in the medulla, and explain which of the three children has lost that process and in which organ. Then a fourth patient has had a splenectomy. Write the three findings from the notes that follow, name the three bacteria that put this patient at risk, and explain from red pulp structure and from marginal zone function why both the red cell findings and the infection risk appear. Finish by writing which single transcription factor defect in thymic medulla produces an autoimmune syndrome and name its three features.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-10-01-C05-BD1

Causes of lymphedema and how it hardens

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C05 Activity: NPLEX-P1-CARD-10-01-C05-BD1 Date: _____

Write the three mechanisms that produce edema by too much filtration, then write what is different about the mechanism of lymphedema and why the fluid that collects is rich in protein. Build a table of the five causes of lymphedema from the notes with one row each, using the columns cause, how drainage fails, and typical setting, and fill every cell. For the parasitic cause, name the organisms, write where the adult worms live and where the microfilariae are found, and write the three things that damage the vessels. For the developmental cause, name the two gene mutations and the chromosomal condition. Then write out the six-step cycle that takes lymphatic outflow falling to further loss of outflow, and beside the fibrosis step name the cells and the growth factor involved. Finish by writing why early lymphedema pits and improves with elevation while late lymphedema does neither, and name the bedside sign at the base of the second toe.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-10-01-C05-BD2

Reason out a milky effusion

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-10 Lymphatic Vessels, Lymphoid Tissues, and Lymphoid Organs Competency: NPLEX-P1-CARD-10-01-C05 Activity: NPLEX-P1-CARD-10-01-C05-BD2 Date: _____

A patient develops a large left pleural effusion after chest surgery. The fluid stays turbid after centrifugation. Write which structure has been injured and, from where the duct lies relative to the midline above and below its crossing, write whether the injury sits above or below that crossing. Then write what the fluid triglyceride would have to be to support this diagnosis, what value makes it unlikely, and which particle confirms it. Write why the fluid was thin and clear while the patient was fasting and turned milky once feeding resumed. Next write how you would separate this fluid from the fluid of an infected pleural space and from a chronic cholesterol effusion, giving the one measurement or observation that settles each. Finish by writing the four things a continuing leak depletes and explaining each loss from what the duct normally carries.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-10-01-C05-CP01

Three kinds of swelling that look alike

Fill in the table from memory, then check it against the notes and correct any cell you missed.

Feature	Lymphedema	Edema of heart failure or low plasma albumin	Myxedema
What accumulates in the tissue			
Protein content of the fluid			
Which Starling force or drainage step has gone wrong			
Pitting or nonpitting, and why			
Response to elevating the limb			
What the skin looks like over time			
Bedside sign or finding that points to it			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-10-01-C05-CA01

A firm, swollen arm four years after breast surgery

A 58-year-old woman comes to the clinic because of swelling of the right arm that began eight months after a right mastectomy with axillary lymph node dissection four years ago. She also had radiation to the right axilla. The swelling was soft at first and went down overnight, but over the past two years it has become firm and no longer changes with elevation. She has had three episodes of a hot, red, tender right arm with fever in the past year. Temperature is 37.1 C (98.8 F), pulse is 78/min, respirations are 14/min, and blood pressure is 128/76 mm Hg. The right arm is larger than the left from the shoulder to the fingers, the skin is thickened and rough, and a fold of skin at the base of the second finger cannot be pinched up. The jugular venous pressure is normal, the heart sounds are normal, serum albumin is 4.1 g/dL (reference range 3.5 to 5.0 g/dL), and the left arm is normal.

1. Recognize. Name the type of edema in this arm and name the two events in her history that caused it, writing for each one how it removed or disabled lymphatic pathways.

2. Explain. Explain why the fluid in this arm is rich in protein, and explain why the arm swells at all when her capillary pressure, plasma albumin, and capillary permeability are normal.

3. Differentiate. Write how this swelling differs from the leg swelling of heart failure and from the swelling of low plasma albumin, naming the findings in this patient that separate them.

4. Investigate. Write which bedside findings and which laboratory values in the case you would use to confirm your answer, and say what each one shows.

5. Decide. State why the arm changed from soft and pitting to firm and nonpitting over two years, and name what is now occupying the tissue.

6. Justify. Justify from the safety factors against edema why losing the nodes and vessels makes this arm swell, and explain from the self-perpetuating cycle in the notes why her repeated episodes of hot, red, tender skin make the swelling worse each time.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-10-01-C04-Q01

Which compartment the organ lacks

A 7-month-old boy has had three episodes of pneumococcal pneumonia and persistent diarrhea. Serum IgG, IgA, and IgM concentrations are all markedly decreased, and the number of circulating T cells is normal. A lymph node biopsy specimen shows a normal paracortex with high endothelial venules but no follicles, no germinal centers, and no plasma cells in the medullary cords. Which of the following best explains these node findings?

A. Failure of thymocytes to be positively selected in the thymic cortex

B. Failure of naive lymphocytes to enter the node through high endothelial venules

C. Arrest of B-cell maturation in the bone marrow

D. Absence of M cells over mucosal lymphoid follicles

Why did you pick it?

Board-style question NPLEX-P1-CARD-10-01-C02-Q02

Where a testicular tumour spreads first

A 29-year-old man has a firm, painless 2-cm mass in the right testis. The overlying scrotal skin is normal. Imaging is done to look for nodal spread. Which group of lymph nodes is most likely to be involved first?

A. Lumbar (para-aortic) nodes near the renal vessels

B. Superficial inguinal nodes

C. Internal iliac nodes

D. Popliteal nodes

Why did you pick it?